1. Initial problem

We consider the heat transfer equation on $\Omega = [0, 2]$:

$$-k(x)u'' = 0$$

$$u'(0) + u(0) = 20$$

$$u(2) = 3$$

$$k(x) = \begin{cases} 1 \text{ for } x \in [0, 1] \\ 2 \text{ for } x \in (1, 2] \end{cases}$$

2. Dirichlet condition

To reduce the non-homogeneous Dirichlet condition to a homogeneous one, we use the substitution $u = w + \overline{u}$. We take the shift function as:

$$\overline{u} = x + 1$$

So that $\overline{u}(2) = 3$, therefore: $w(2) = 0$. The derivative of the shift is $\overline{u}' = 1$. In summary:

$$u = w + x + 1$$

$$u' = w' + 1$$

$$u'' = w''$$

3. Robin condition

Using the shift, we transform the condition $u'(0) + u(0) = 20$:

$$w'(0) + 1 + w(0) + 1 = 20$$

$$w'(0) + w(0) = 18$$

$$w'(0) = 18 - w(0)$$

4. Weak formulation

We multiply the equation by a test function $v : \forall v \in V v(2) = 0$ and integrate by parts:

$$\int_0^2 k(x)u'v' \ \mathrm{d}x - [k(x)u'v]_0^2 = 0$$

Substituting $u = w + \overline{u}$ and expanding the boundary term at $x = 0$ (keeping in mind that $v(2) = 0$):

$$\int_0^2 k(x)w'v' \ \mathrm{d}x + \int_0^2 k(x)v' \ \mathrm{d}x + k(0)w'(0)v(0) + k(0)v(0) = 0$$

Substituting the Robin condition into the variational equation with $k(0) = 1$:

$$\int_0^2 k(x)w'v' \ \mathrm{d}x + \int_0^2 k(x)v' \ \mathrm{d}x + (18 - w(0))v(0) + v(0) = 0$$

$$\int_0^2 k(x)w'v' \ \mathrm{d}x - w(0)v(0) = -19v(0) - \int_0^2 k(x)v' \ \mathrm{d}x$$

5. Final form $B(w, v) = L(v)$

$$B(w,v) = \int_0^2 k(x) w' v' \, dx - w(0)v(0)$$

$$L(v) = -19v(0) - \int_0^2 k(x) v' \, dx$$